LaTeX SupCon-guided Hybrid Classification for Tail-aware Representation Learning

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1. Introduction

Long-tailed image classification is a crucial challenge in modern visual recognition tasks, since there is a need to address the unbalanced class distributions inherent in massive-scale datasets. In such cases, head classes which are a smaller number of classes dominate the gradient during loss backpropagation causing models to primarily generalize toward these head classes. As a result, accuracy on the tail classes remains relatively poor. To tackle this problem various approaches have emerged, Zhang et al. [11] classified these methods into Class Re-balancing, Information Augmentation and Module Improvement.

Class Re-balancing approaches aim to mitigate the imbalance in datasets by modifying sampling probabilities or adjusting logits. The most common methods are random over-sampling (ROS) and Random Under-Sampling (RUS). ROS samples more data at tail classes and RUS undersamples at head classes to balance the number of data between head and tail. Recently, advanced sampling strategy like square-root sampling[7] and progressively balanced sampling[4] are usually utilized. Moreover, logit adjustments such as re-weighting or margin adjustment improves generalization quality at tail classes.

Information Augmentation methods try to improve the representation of tail classes by feature transformations and data augmentations. This approach includes basic augmentations like random flipping or cropping and advanced techniques such as embedding mix-up[6] and contrastive learning paradigms[8]. These strengthen the robustness of tail classes and make them distinguishable.

Module Improvement approaches enhance specific modules or components within neural architecture. These often include designing specific architectures, improving classifier structures or employing two-stage trainers which are called decoupled training[4].

Among these Supervised Contrastive Learning (SupCon)[5] has showed prominent results by encouraging encoder to learn discriminative feature, enhancing inter class separability and intra class compactness. SupCon yields a **Plug-and-Play Representation** that any downstream classifier can attach to without additional feature re-engineering since SupCon already provide general encoder which has prominent feature.

To exploit this advantage, we plug a hybrid classifier into the SupCon-guided encoder. In this work we attach only an LDAM[1] classifier and a Cross-entropy(CSE) classifier but any other classifier could be leveraged. SupCon is complementary to LDAM: SupCon derives the encoder toward lower intra-class variance and higher inter-class distance, which enables LDAM and CSE to make more effective and robust decision boundaries.

Finally, since top-1 accuracy can mask class-level behavior, we report relative accuracy[11] and head-middle-tail accuracy breakdown to provide a clearer performance evaluation results. Followings summarized our key contributions:

1. Two-Stage Training Pipeline with SupCon Guidance
We propose a two-stage pipeline training framework for long-tailed distribution classification, leveraging SupCon as first stage to guide feature learning. This strategy enables enhancing both head and tail class accuracy by integrating encoder and classifiers which trained separately.

2. Plug-and-Play Encoder with Hybrid Classifiers
Using the plug-and-play property of the SupCon encoder, we decouple feature learning and classifier fine-tuning. We attach two complementary classifiers LDAM which outperforms in tail classes and CSE in head classes. This strategy enhances robustness while preserving compatibility with previously trained model.

3. Various Ensemble methods
We utilized various ensemble methods, confidence-based

prediction and fixed softgate logits. By this strategies a model is able to mix different classifiers properly and enhance the performance of the model.
Our code is available at: https://github.com/jeoncwcw/DeepLab_final_project

2. Related Works

Long-tail recognition, a significant challenge in computer vision, tackling imbalanced datasets, has difficulty at its skewed distribution. Previous approaches often failed to balance accuracy between dominant head classes and minority tail classes, leading to biased predictions for majorities. Various methodologies have emerged to solve this problem, such as Decoupled Training and contrastive learning.

Decoupled Training
Decoupled Training methods split the learning process into distinct two phases: representation learning and classifier fine-tuning. Kang et al.[4] demonstrated that while class-balanced re-weighting benefits classifier training, it negatively impacts feature representation. To solve this trade-off, Kang et al.[4] suggested that firstly training a feature representation and subsequently fine-tuning classifiers with a re-sampled balanced dataset.

Supervised Contrastive Learning
SupCon which presented by Khosla et al.[5] utilizes label data to enhance intra-class compactness and separate inter-class in feature space. Previous works[2] combined SupCon with CSE. Specifically Hybrid-SC[8] demonstrated the effectiveness of end-to-end training using SupCon and CE, significantly outperforming traditional ways.

Integration of SupCon and LDAM
Recent studies have explored combining SupCon and LDAM to leverage their complementary strengths. They have illustrated that pairing these losses can produce balanced accuracy improvements across class distributions. Especially, Wang et al.[9] introduce joint-learning between LDAM and SupCon and marginally outperform base LDAM.

Novelty and Superiority of Our Strategies
Unlike prior works that focus on either representation learning or classifier design, our method jointly leverages SupCon for encoder and a hybrid classifier of LDAM and CSE. This pipeline exploits Supcon's plug-and-play nature. Moreover we apply various methods especially relative accuracy to validate our model.

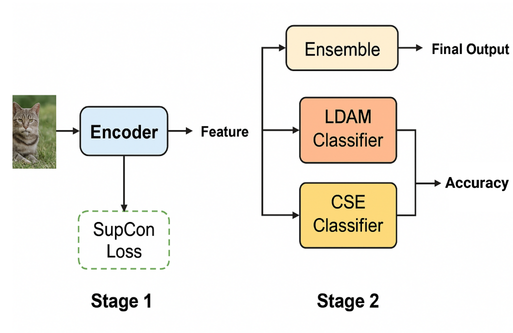

Figure 1. Overview of our proposed two-stage architecture for tail-aware representation learning. Stage 1 employs a Supervised Contrastive Loss (SupCon) to guide the encoder toward extracting discriminative features. In Stage 2, the learned encoder features are leveraged by complementary classifiers (LDAM and Cross-Entropy) and finetuned these classifiers. When testing the model, classifiers are combined via ensemble methods to enhance balanced performance across head and tail classes.

3. Approach

3.1. LDAM and SupCon

In long-tailed classification settings, models tend to perform well on head classes but poorly on tail classes due to the significant imbalance in class frequencies. This imbalance causes gradients to be dominated by the overrepresented head classes and leading to biased decision boundaries. To mitigate this issue, the Label-Distribution-Aware-Margin(LDAM) loss [1] introduces a new class rebalancing strategy. Its idea is quite simple, give a larger margin at low-frequency classes, to help them occupy a larger region in the feature space by adjusting decision boundaries. This enhances tail classes discriminability. The LDAM Loss is defined as follows:

$$L_{LDAM}(x, y) = -\log \frac{e^{z_y - \Delta_y}}{e^{z_y - \Delta_y} + \sum_{j \neq y} e^{z_j}}, \tag{1}$$

where x is model input, y is label, j is the class index and z_j is the logit corresponding to the class j. LDAM is based on Cross-Entropy loss but differs by utilizing Δ_y. Δ_y is usually calculated as follows (in some data set, they can be modified):

$$\Delta_y = \frac{C}{n_y^{1/4}} \text{ where } n_y \text{ : number of examples at class y} \tag{2}$$

C is hyper parameter to adjust value of $\Delta_y . n_y$ is inversely related with Δ_y, so Δ_y becomes greater when number of examples decrease. Since the difference between the number of examples in the largest class and the smallest class in long-tailed dataset is very big, Δ_y becomes much greater at tail classes even with power $\frac{1}{4}$. When Δ_y becomes greater, it makes a term $e^{z_y - \Delta_y}$ becomes smaller. So, numerator at Equation 1 becomes smaller since LDAM loss is negative log probability loss for tail classes become relatively larger than head classes. Consequently, Δ_y works as margin in LDAM loss and leads tail classes to have larger margin and higher loss so model tries to adjust parameters harder for tail class examples.

However, when LDAM is used in conjunction with aggressive data augmentations such as random flipping, cropping, or color jitter the features of augmented examples may significantly deviate from their class' center. This deviation can result in inconsistency and disturb optimizer to draw proper decision boundaries which have enough margin for tail classes, thereby degrading performance.

To alleviate this problem, many methods can be applied. One example is adding an L1 loss term to minimize distance between augmented data and its original data. This method is applied at crGAN[10] to minimizing distance between same classes. But, due to ease of implementation and explainability, SupCon[5] is major technique to reduce variation at intra-class and increase distance at inter-class.

This stabilizes the feature space and helps LDAM converge more effectively. SupCon loss is formulated as:

$$L_{SupCon} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\sum_{p\in P(i)} \exp(z_i \cdot z_p/\tau)}{\sum_{a\in A(i)} \exp(z_i \cdot z_a/\tau)}, \quad (3)$$

where N is the number of sample, P(i) denotes the set of positive samples for anchor i, A(i) is the set of all samples, z represents normalized embeddings where i means current sample, p for positive samples, τ for hyperparameter and a for all other samples. By guiding the encoder to learn well-structured features, SupCon provides a solid foundation that mitigates the negative impacts of data augmentation on LDAM margin design.

3.2. Model Architecture

As shown in Figure 1 our model consists of two-stages. First stage is for encoder (or backbone), second stage is for classifiers. Thanks to the plug-and-play feature of SupCon, it is able to adopt various classifiers to a single encoder. We implement CSE and LDAM classifiers, but any other classifiers can be attached at the end of model. To utilize this strength we implemented various ensemble methods described in Section 3.3

Our model adopts ResNet-18[3] as the backbone due to its simplicity of implementation and lack of our computational resources. Common ResNet-18 is produced and trained to solve classification problem on the ImageNet whose dataset has high-resolution images (224 $\times$ 224). However, we try to tackle classification at CIFAR-100 which has low-resolution (32 $\times$ 32). This causes critical problem, feature spaces at CNN layer become so small and makes it harder to learn its representation. So, we modify the standard ResNet-18 by removing the max-pooling layer and changing kernel size to 3 $\times$3. This ensures that spatial resolution is preserved in early stages of the network.

To effectively utilize the LDAM loss, we replace the standard classification layer to Normalized classification layer at the LDAM classifier. Since LDAM is logit margin-based loss, we need to control its scale. So, we normalize and scale by adopting scaling factor(s) set to 30. Furthermore, we integrate a projection head composed of two fully connected layers on top of encoder specifically for computing SupCon loss. This projection head is used exclusively during training encoder with SupCon and discarded during classifiers are fine-tuned. It allows the encoder to learn embeddings optimized for contrastive objectives without compromising the classification pathway.

3.3. Ensemble Methods

To fully leverage the hybrid classifier which has complementary strengths of the LDAM and CSE classifiers, we introduce two ensemble strategies: the Fixed Softgate Logits and the Confidence-Based Prediction. These methods enable the model to adaptively or statically combine classifier outputs to improve robustness and accuracy.

Fixed Softgate Logits methods combines the logits of the two classifiers using a fixed linear parameter:

$$z_{\text{ensemble}} = \alpha \cdot z_{\text{LDAM}} + (1-\alpha) \cdot z_{\text{CSE}}, \quad (4)$$

where z is the classifier output. We initialize and fix α as follows:

$$\alpha_i = \frac{\sqrt{n_{\text{head}}}}{\sqrt{n_i} + \sqrt{n_{\text{head}}}} \quad (5)$$

where n_{head} means the number of the largest class and n_i is the number of class at index i. When n_i decreases, the portion of numerator becomes larger and α_i increases. Conversely, when n_i increases, α_i decreases. Consequently, LDAM has a larger influence on tail class and smaller at head. Moreover, α becomes $\frac{1}{2}$ at n_{head} so following equation is valid:

$$\alpha_i \geq \frac{1}{2} \quad (6)$$

This means the proportion of LDAM is always same or bigger than CSE even in head classes. This design reflects our theoretical assumption that LDAM provides more stable and generalizable decision boundaries, especially under class-imbalance settings. By ensuring that LDAM maintains at least equal influence, the ensemble avoids the potential risk of overfitting to head classes, which often occurs with purely weight-based rebalancing methods like CSE. Moreover, the fixed gating mechanism offers implementation simplicity, while still leveraging the respective strengths of both classifiers.

Confidence-Based Prediction strategy dynamically selects the output of the classifier with the higher confidence score for each individual input:

$$z_{\text{ensemble}}(x) = \begin{cases} z_{\text{LDAM}}(x), & \text{if } \max(z_{\text{LDAM}}(x)) \geq \max(z_{\text{CSE}}(x)) \\ z_{\text{CSE}}(x), & \text{otherwise} \end{cases} \tag{7}$$

This method improves decision reliability by selecting the classifier that exhibits stronger confidence on a per-sample basis. The trade-off is a slight increase in computational complexity due to additional condition checks and maximum operations. Nevertheless, the added adaptability often results in more robust and accurate predictions, especially with highly varied class distributions.

4. Experiments

4.1. Implementation Details

We divide models as two types: Single-Stage models and Multi-Stage models to fully understand pros and cons of diverse strategies. To evaluate models properly, balanced and unbalanced training data used in order to analyze impact of class imbalance on model performance. The imbalanced dataset was constructed by sampling the training data of CIFAR-100, where the most frequent class contains 450 samples and the least frequent contains 54 samples. The number of samples for the remaining classes was linearly interpolated between these two extremes.

Single-Stage Models

Models that are fully trained by CSE loss or LDAM loss are implemented as Single-Stage Models and training strategy follows Zhang et al. [11], Wang et al. [9]. Both of CES model and LDAM model utilize SGD optimizer, with learning rate of 0.1 for LDAM and 0.025 for SCE. Momentum and weight decay are fixed at 0.9 and 0.0005. Specifically, the Deferred Re-weighting (DRW) strategy[1] is applied to improve performance by assigning greater weight to tail classes.

Different scheduling strategies are applied to CSE and LDAM models. For CSE model, learning rate linearly decays from initial value to zero according to the following equation:

$$\eta_t = \eta_0 \left(1 - \frac{t}{T}\right) \tag{8}$$

For LDAM model, cosine annealing with warmup epochs strategy is used by following equation:

$$\eta_t = \begin{cases} \eta_0 \cdot \frac{t+1}{T_{\text{warm}}}, & \text{if } t < T_{\text{warm}} \\ \eta_0 \cdot \frac{1}{2}\left(1 + \cos\left(\pi \cdot \frac{t - T_{\text{warm}}}{T - T_{\text{warm}}}\right)\right), & \text{if } t \geq T_{\text{warm}} \end{cases} \tag{9}$$

A large margin for tail classes and a scaling factor applied at LDAM loss can cause gradients to become unstable and large. This excessive margin disturbs the formation of robust representation at the early stage of training. So, to mitigate this issue, a warm-up strategy is needed which helps stabilize the learning process. Moreover, the smooth nature of cosine annealing helps tuning decision boundary finely.

We train the Single-Stage Models at 100 epochs and find models converge well.

Algorithm 1 Three-Phase Training Strategy for Multi-Stage Model

Model Components: Encoder E, Projection Head P, LDAM Classifier C_L, CSE Classifier C_C

Phase 1: SupCon Fine-tuning (100 epochs)
  Train: Encoder E (using SupCon loss)
  Freeze: C_L, C_C
  Used: Projection Head P to map to SupCon space

Phase 2: LDAM Classifier Fine-tuning (60 epochs)
  Train: LDAM Classifier C_L (using LDAM loss)
  Freeze: E, P, C_C

Phase 3: CSE Classifier Fine-tuning (60 epochs)
  Preprocess: Apply Random Under Sampling (RUS) to balance class distribution
  Train: CSE Classifier C_C (using Cross-Entropy loss)
  Freeze: E, P, C_L

The Multi-Stage Model

Algorithm 1 shows the details of the training pipeline applied to Multi-Stage Model. In phase 1, the encoder and the projection layer are trained by SupCon loss while other layers are frozen. Similar to LDAM warmup and cosine annealing strategies are applied. During this process, the encoder enhances its feature representation and a decision boundary without interference from CSE and LDAM classifiers. We set the batchsize to 256 in dataloader to allow SupCon to observe a large number of examples.

In phase 2, the LDAM classification layer is trained by LDAM loss while others are frozen. Warmup and cosine annealing are leveraged again. With help of well-represented encoder, classifier converges quickly.

Lastly, in phase 3, before fine-tuning the CSE classification layer, we apply RUS on unbalanced data since the CSE classifier performs better on balanced datasets. Moreover, the linear decay learning rate schedule is applied. Please refer to the code for additional implementations.

4.2. Experimental Results

A simple Top-1 Accuracy is not enough to represent model's performance and ability. Especially, when dealing with unstructured datasets such as long-tail distributions, the reliability of Top-1 accuracy significantly decreases.

Table 1. Top-1 Accuracy and Head-Middle-Tail accuracy (%) for Each Ensemble Strategy across Class Splits at the Multi Stage Model

Class Split	LDAM	CSE	Confidence	Softgate
Overall (Top-1)	64.40	62.40	64.37	**64.46**
Head	71.20	**72.20**	71.80	71.30
Middle	62.40	60.00	61.80	**62.60**
Tail	56.10	51.20	56.20	**56.50**

Table 2. Top-1 Accuracy and Head-Middle-Tail accuracy (%) Comparison between Single-Stage (Unbalanced dataset) and Multi-Stage Models (Softgate Strategy)

Class Split	CSE	LDAM	Multi-Stage
Overall (Top-1)	60.49	61.17	**64.46**
Head	70.36	65.48	**71.30**
Middle	61.62	61.12	**62.60**
Tail	49.45	**56.91**	56.50

Therefore, to properly evaluate models on long-tail datasets, we employed Head-Middle-Tail accuracy and relative accuracy[11] alongside Top-1 accuracy, allowing for a more comprehensive and multi-faceted assessment of model performance.

Head-Middle-Tail Accuracy

The idea of Head-Middle-Tail accuracy is quite simple: just measure Top-1 accuracy of head, middle and tail classes. We divide these classes by selecting 33%, 34%, and 33% of the full train data as the head, middle, and tail sets, respectively. We divide the test and validation datasets using the same methodology.

By applying Head-Middle-Tail accuracy along with Top-1 accuracy we conduct a quantitative evaluation of our models. First, as shown in Table 1, we compare four different ensemble strategies. Two of them are the outputs of individual classifiers: the CSE classifier and the LDAM classifier. The other two are the Fixed Softgate Logits and the Confidence-Based Prediction methods, which are described in Section 3.3.

Among the four strategies, the Softgate method achieves the highest overall Top-1 accuracy (64.46%), outperforming both individual classifiers and the other ensemble method. Although the CSE classifier achieves the highest performance on head classes (72.2%), it reports relatively worse result at the other accuracies. Especially at the tail classes CSE shows significantly poor performance (51.20%), indicating poor generalization across imbalanced data. On the other hand, the LDAM classifier performs more consistently across distribution, especially improving tail accuracy (56.10%) compared to CSE.

Notably, both ensemble strategies demonstrate robust performance among all splits. Even though CSE classifier's performance collapsed on the tail classes, Softgate and Confidence strategy outperform LDAM by combining their outputs. These results support the effectiveness of ensemble strategies in long-tailed scenarios and validate that fusion methods can improve both overall and class-specific performance.

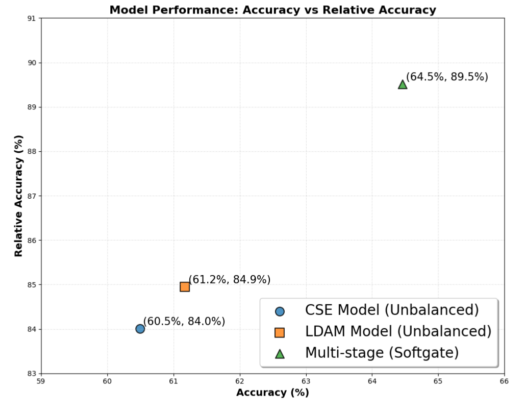

Figure 2. Comparison of Top-1 Accuracy and Relative Accuracy (%) across different models. The CSE and LDAM single-stage models achieve relatively low relative accuracy, indicating poor adaptation to class imbalance. In contrast, the proposed Multi-Stage model with the Softgate ensemble strategy achieves the highest relative accuracy (89.5%) while also maintaining superior overall accuracy (64.5%), demonstrating robust performance across both head and tail classes in long-tailed distribution

Secondly, we compare the Multi-Stage Model with the Single-Stage models, applying Softgate strategy to the Multi-Stage Model. As shown in Table 2, the Multi-Stage Model outperforms the Single-Stage models by a margin of 3-4%. In detail, the Multi-Stage shows better results across all categories except for the tail classes. The LDAM model achieves the best performance on tail classes, but at the cost of reduced accuracy on head classes. In contrast, the Multi-Stage Model achieves comparable performance on tail classes while outperforming the CSE model on head classes. These results demonstrate that the Multi-Stage Model effectively compensate for tail class representation while maintaining strong performance on head classes.

Relative Accuracy

Relative accuracy provides a more balanced evaluation metric under long-tailed distributions by comparing model performance on tail classes relative to a balanced baseline. This highlights improvements in minority class recognition that are often overlooked by overall accuracy metrics. Cao et al. [1] define Relative Accuracy A_r as the ratio between test accuracy under long-tailed training(A_t) and the best test accuracy achieved from balanced training settings divided as balanced data with standard Training(A_v) and balanced data with long-tail-specific training(A_u). Finally Relative Accuracy is formulated as follows:

$$A_u = \max(A_v, A_b) \tag{10}$$

$$A_r = \frac{A_t}{A_u} \tag{11}$$

Figure 2 plots the Top-1 accuracy and Relative accuracy for the Single-Stage models with pretrained at unbalanced dataset and the Multi-Stage model with Softgate ensemble method. Due to the time constraints of training with a balanced dataset, the Multi-Stage model's relative accuracy was computed using only A_v instead of A_u. As shown in the figure, The Multi-Stage model achieves the highest performance in both relative accuracy and Top-1 accuracy.

Qualitative Results

Another notable characteristic of the Multi-Stage model is its flexibility in fixing, fine-tuning, or adding classifiers. Its Plug-and-Play feature described in Section 1, and the discriminative representations enable those strengths. To quantify this, we measure the ratio of inter-class distance to intra-class variance at the end of the encoder. The Multi-Stage model achieves a ratio of 2.2 at the end of the training, indicating a high degree of class separability. This discriminative property also contributes to faster convergence. In our experiments, classifiers in the Multi-Stage model converged within approximately 10 epochs (about 10 minutes), whereas the Single-Stage model required nearly 2 hours to reach convergence. Therefore, the proposed Multi-Stage model offers the advantage of easy maintenance and improvement, as long as the encoder is pre-trained. Moreover, all inference or evaluate process proposed in this paper does not take more than one minute.

5. Conclusion

In this paper, we proposed a novel Multi-Stage Model that addresses the long-tailed distribution problem by incorporating Supervised Contrastive Learning (SupCon) into the feature learning phase. The proposed model consists of a shared encoder pre-trained using SupCon loss, followed by two specialized classifiers, one trained with LDAM loss and the other with Cross-Entropy (CSE) loss.

Compared to existing approaches such as LDAM[1], a modern method for long-tailed classification, the Multi-Stage Model consistently achieves superior performance, not only in overall Top-1 accuracy but also across Head-Middle-Tail class splits. In particular, it maintains strong accuracy on tail classes without sacrificing performance on head classes, a common limitation in long-tailed scenarios.

While SupCon may not directly optimize the classification objective, it significantly improves the quality of feature representations. The encoder trained with SupCon creates discriminative and compact embeddings, leading to faster classifier convergence and higher generalization. Furthermore, this structure enables Plug-and-Play flexibility, allowing new classifiers to be easily added or fine-tuned without retaining the entire model.

In summary, the Multi-Stage Model is not only effective in addressing long-tail distribution but also offers practical advantages in terms of training efficiency, scalability, and modularity. This makes it a strong candidate for real-world long-tailed recognition tasks.

However, there could be some future extensions for this architecture. The Fixed Softgate Logits, the ensemble method applied in this model, could be improved by parameterizing and training it, initialized with the value proposed in Sec 3.3. Moreover, it would not incur significant computational cost due to its simplicity.

Adding various types of classifiers and measuring their relative accuracy along with Top-1 accuracy will be also helpful in further improving the model. While the relative accuracy and Top-1 accuracy shown in Figure 2 exhibit only a linear relationship, increasing the number of classifier types may provide a more robust evaluation metric for models under long-tailed distributions.

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